

Secure Self-Hosted File Collaboration Platform on Azure

Project Author: Esmeralda C. Cabrera Ventura

Program: PGP in Cloud Computing

Platform: Microsoft Azure

Project Overview

This project implements a **secure, self-hosted cloud file sharing platform** using **OwnCloud** deployed in a controlled Azure environment. The solution addresses enterprise concerns around employees using unauthorized public file-sharing services such as Dropbox, which can expose confidential data outside corporate governance and increase breach risk. Businesses must prevent confidential data leakage while still enabling collaboration.

The architecture ensures:

- Data remains inside an organization-controlled cloud boundary.
- Secure separation between application and database layers.
- Restricted internet exposure using private networking.
- Compliance with internal security and governance policies.

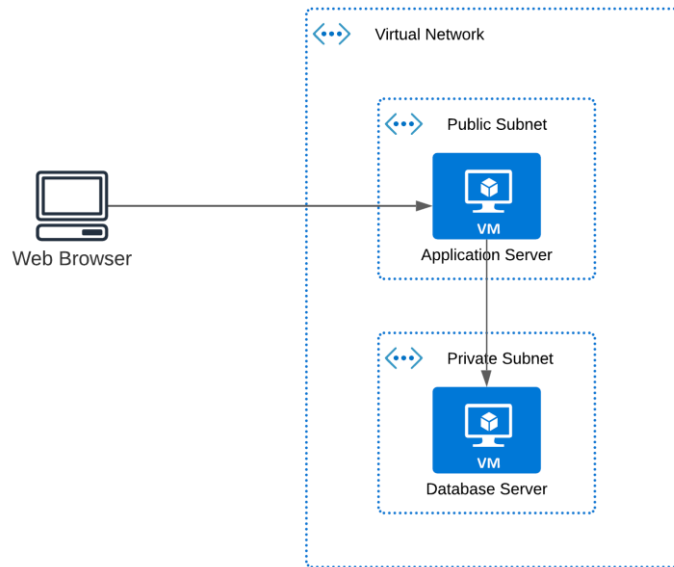
Architecture Summary

The solution deploys:

- A **Virtual Network (VNet)** with segmented public and private subnets.
- An **Application Server (OwnCloud)** hosted in the public subnet.
- A **Database Server (MySQL)** hosted in the private subnet.
- A **NAT Gateway** to allow secure outbound access from the private subnet.
- A **Bastion-style SSH access model** (jumping through the application VM).
- **Network Security Groups (NSGs)** enforcing least-privilege access.

This design follows a classic **3-tier secure cloud pattern**.

Architecture Diagram



Project Objectives

The architecture implementation required:

1. Create separate public and private subnets within a virtual network.
2. Install MySQL on a VM located in the private subnet.
3. Install OwnCloud on a VM located in the public subnet.
4. Configure NSGs to allow only required ports.
5. Validate deployment through browser access to the application server.

Technologies Used

Category	Technology
Cloud Platform	Microsoft Azure
Compute	Azure Virtual Machines (Ubuntu)
Networking	Azure VNet, NAT Gateway, NSGs
Application	OwnCloud
Database	MySQL
Security	SSH Key Authentication
Web Stack	Apache + PHP
OS	Ubuntu 22.04 LTS

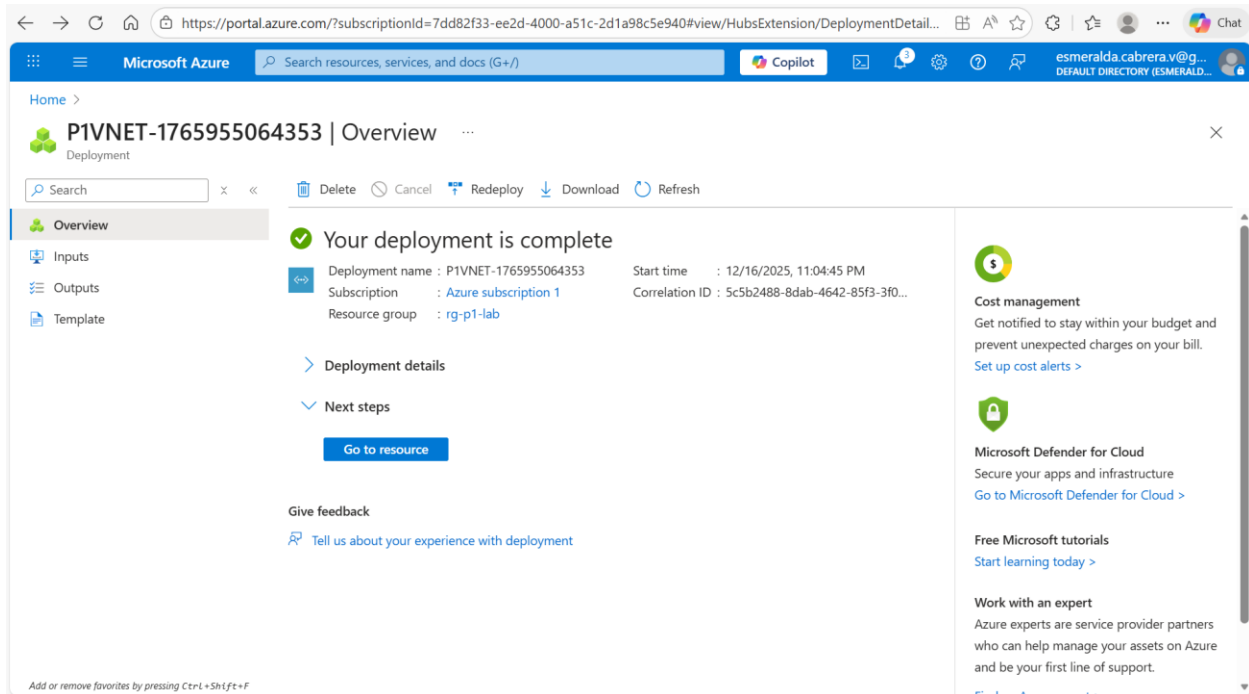
Step-by-Step Implementation

Step 1: Resource Group Creation

All project resources were deployed inside a single Azure Resource Group to maintain lifecycle control and simplify cleanup.

Actions: Virtual Network and Subnet Creation

Virtual Network Deployment



Home > P1VNET-1765955064353 | Overview

Deployment

Search

Delete Cancel Redeploy Download Refresh

Overview

Inputs

Outputs

Template

Your deployment is complete

Deployment name : P1VNET-1765955064353 Start time : 12/16/2025, 11:04:45 PM

Subscription : Azure subscription 1 Correlation ID : 5c5b2488-8dab-4642-85f3-3f0...

Resource group : rg-p1-lab

Deployment details

Next steps

Go to resource

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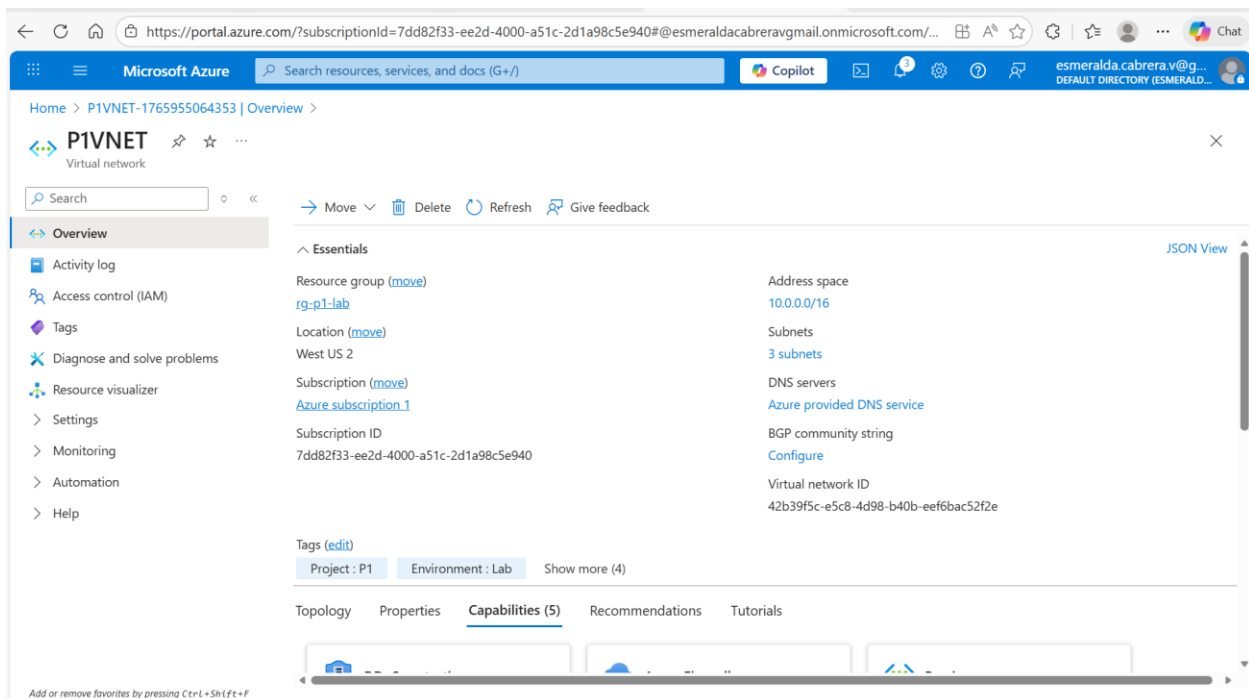
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VNET Creation with Properties



Home > P1VNET-1765955064353 | Overview

P1VNET Virtual network

Search

Move Delete Refresh Give feedback

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

Settings

Monitoring

Automation

Help

Essentials

Resource group (move) rg-p1-lab

Location (move) West US 2

Subscription (move) Azure subscription 1

Subscription ID 7dd82f33-ee2d-4000-a51c-2d1a98c5e940

Address space 10.0.0.0/16

Subnets 3 subnets

DNS servers Azure provided DNS service

BGP community string Configure

Virtual network ID 42b39f5c-e5c8-4d98-b40b-eef6bac52f2e

Tags (edit)

Project : P1 Environment : Lab Show more (4)

Topology Properties Capabilities (5) Recommendations Tutorials

List of VNET Subnets

The screenshot shows the Microsoft Azure portal interface. The top navigation bar includes the Microsoft Azure logo, a search bar, and the user profile 'esmeralda.cabrera.v@g...'. The breadcrumb trail indicates the path: Home > P1VNET-1765955064353 | Overview > P1VNET. The main heading is 'P1VNET | Subnets' with a star icon and a close button. A left-hand navigation pane lists various services, with 'Subnets' highlighted. The main content area displays a table of subnets with columns: Name, IPv4, IPv6, Available IPs, Delegated to, Security group, and Route table. Below the table, it states 'Showing 3 subnets' and provides a 'Give feedback' link.

Name	IPv4	IPv6	Available IPs	Delegated to	Security group	Route table
PublicSubnet	10.0.1.0/24	-	251	-	-	-
PrivateSubnet	10.0.2.0/24	-	251	-	-	-
AzureBastionSubnet	10.0.255.0/26	-	59	-	-	-

Public Subnet Properties

The screenshot shows the 'Edit subnet' dialog in the Microsoft Azure portal. The dialog is titled 'Edit subnet' and contains the following fields and options:

- Subnet ID:** /subscriptions/7dd82f33-ee2d-4000-a51c-2d1a98c5e940/resourceGroups/rg-p1-lab/provi...
- Subnet purpose:** Default
- Name:** PublicSubnet
- IPv4:**
 - Include an IPv4 address space:** ☒
 - Choose a starting address and size within your IPv4 address range:** 10.0.0.0 - 10.0.255.255
 - Starting address:** 10.0.1.0
 - Size:** /24 (256 addresses)
 - Subnet address range:** 10.0.1.0 - 10.0.1.255
- IPv6:**
 - Include an IPv6 address space:** ☐ This virtual network has no IPv6 address ranges.
- Private subnet:** (Section header)
- Buttons:** Save, Cancel

The background shows the 'P1VNET | Subnets' page with the 'PublicSubnet' selected in the list.

Private Subnet Properties

Microsoft Azure

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Copilot

esmeralda.cabrera.v@g...
DEFAULT DIRECTORY (ESMERALDA...)

Home > PIVNET-1765955064353 | Overview > PIVNET

PIVNET | Subnets

Virtual network

Search

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

Settings

Address space

Connected devices

Subnets

Bastion

DDoS protection

Firewall

Microsoft Defender for Cloud

Create subnets to segment the resource an IP address

Search subnets

Name ↑

PublicSubnet

PrivateSubnet

AzureBastion

Showing 3 subnets

Add or remove favorites by pressing Ctrl+Shift+F

Edit subnet

Select an address space and configure your subnet. You can customize a default subnet or select from subnet templates if you plan to add select services later. [Learn more](#)

Subnet ID: /subscriptions/7dd82f33-ee2d-4000-a51c-2d1a98c5e940/resourceGroups/rg-p1-lab/provi...

Subnet purpose: Default

Name: PrivateSubnet

IPv4

Include an IPv4 address space: ☒

Choose a starting address and size within your IPv4 address range: 10.0.0.0/16 (10.0.0.0 - 10.0.255.255)

Starting address: 10.0.2.0

Size: /24 (256 addresses)

Subnet address range: 10.0.2.0 - 10.0.2.255

IPv6

Include an IPv6 address space: ☐ This virtual network has no IPv6 address ranges.

Private subnet

Save Cancel

Give feedback

Azure Bastion Subnet Properties

Microsoft Azure

Search resources, services, and docs (G+)

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Home > PIVNET-1765955064353 | Overview > PIVNET

PIVNET | Subnets

Virtual network

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Settings

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Connected devices

Subnets

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Create subnets to segment the resource an IP address

Search subnets

Name ↑

PublicSubnet

PrivateSubnet

AzureBastion

Showing 3 subnets

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Edit subnet

Select an address space and configure your subnet. You can customize a default subnet or select from subnet templates if you plan to add select services later. [Learn more](#)

Subnet ID: /subscriptions/7dd82f33-ee2d-4000-a51c-2d1a98c5e940/resourceGroups/rg-p1-lab/provi...

Subnet purpose: Azure Bastion

Name: AzureBastionSubnet

IPv4

Include an IPv4 address space: ☒

Choose a starting address and size within your IPv4 address range: 10.0.0.0/16 (10.0.0.0 - 10.0.255.255)

Starting address: 10.0.255.0

Size: /26 (64 addresses)

Subnet address range: 10.0.255.0 - 10.0.255.63

IPv6

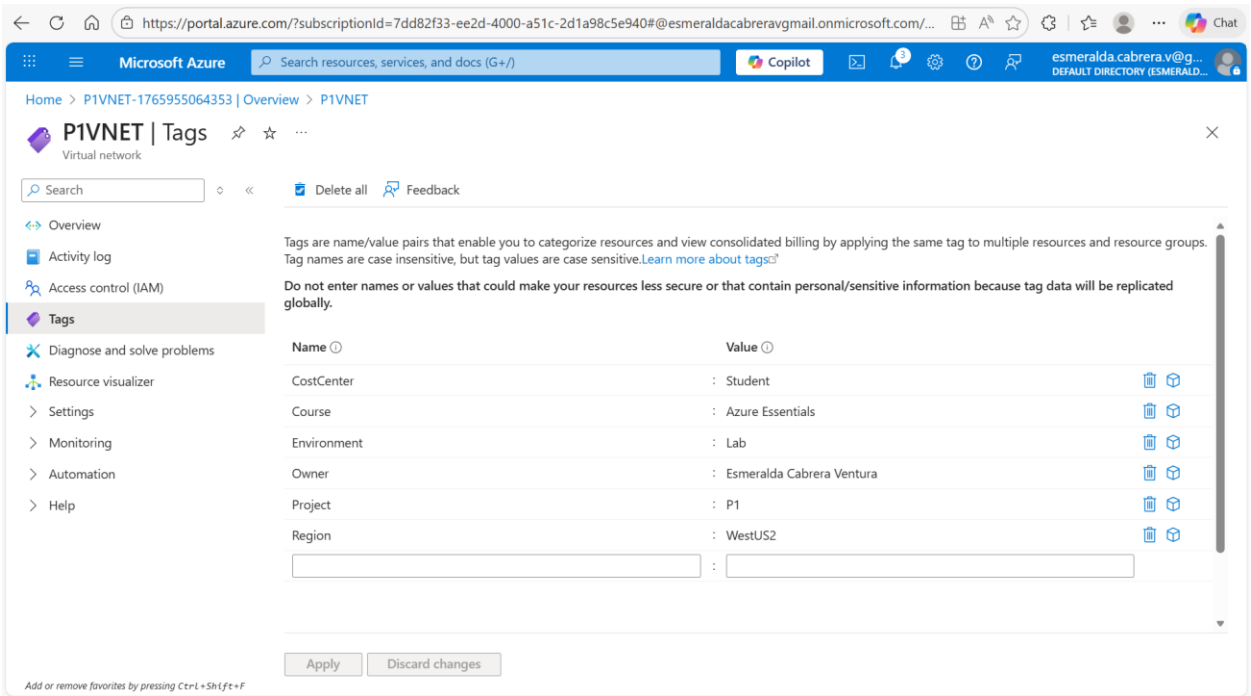
Include an IPv6 address space: ☐ This virtual network has no IPv6 address ranges.

Private subnet

Save Cancel

Give feedback

Tags Used



Step 2: Virtual Network Design

A virtual network named **P1VNET** was created using CIDR 10.0.0.0/16.

Subnets Created

Subnet	CIDR	Purpose
Public Subnet	10.0.1.0/24	Hosts OwnCloud Web Server
Private Subnet	10.0.2.0/24	Hosts MySQL Database Server

This segmentation enforces security isolation between the web tier and data tier.

NAT Gateway — Outbound IP Configuration

← ↻ 🏠 https://portal.azure.com/?subscriptionId=7dd82f33-ee2d-4000-a51c-2d1a98c5e940#view/Microsoft_Azure_Network/NatGateway...

Microsoft Azure Search resources, services, and docs (G+/) Copilot esmeralda.cabrera.v@g... DEFAULT DIRECTORY (ESMERALD...

Home > Network foundation | NAT gateways >

Create network address translation (NAT) gateway

Basics **Outbound IP** Networking Tags Review + create

StandardV2 SKU in the Basics tab.

+ Add public IP addresses or prefixes

Public IP addresses

Name ↕	IP address ↕	IP version ↑	Remove
(new) pip-nat-p1		IPv4	×

Public IP prefixes

Name ↕	IP prefix ↕	IP version ↕	Remove
(new) pip-prefix-p1		IPv4	×

Previous Next **Review + create** Give feedback

NAT Gateway Subnet Association

← ↻ 🏠 https://portal.azure.com/?subscriptionId=7dd82f33-ee2d-4000-a51c-2d1a98c5e940#view/Microsoft_Azure_Network/NatGateway...

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Home > Network foundation | NAT gateways >

Create network address translation (NAT) gateway

Basics Outbound IP **Networking** Tags Review + create

To use the NAT gateway, select a virtual network and then select a subnet within that network. [Learn more](#)

Virtual network P1VNET

Select specific subnets PrivateSubnet (10.0.2.0/24)

Previous Next **Review + create** Give feedback

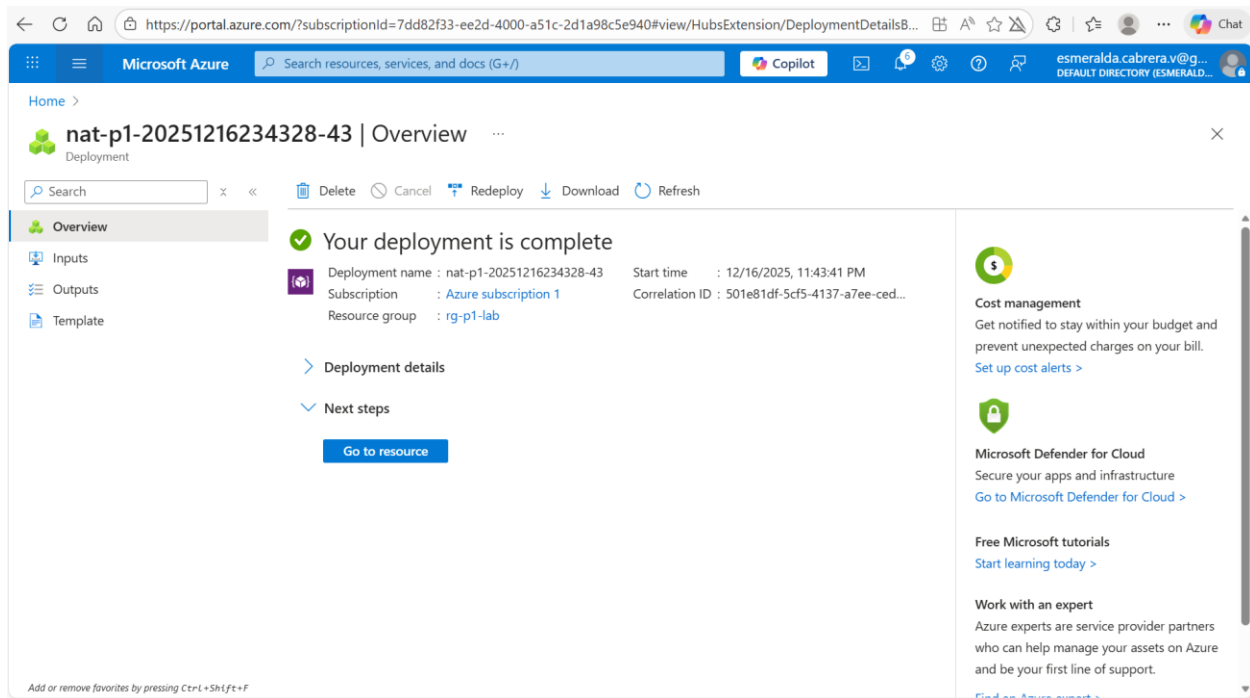
Step 3: NAT Gateway Deployment

A NAT Gateway was attached to the **private subnet** to allow outbound internet access without exposing the database to inbound traffic.

Why This Matters

- Enables patching and package installation.
- Prevents direct public IP exposure of the database.

NAT Gateway Deployment



The screenshot displays the Microsoft Azure portal interface. The top navigation bar includes the Microsoft Azure logo, a search bar, and user information for 'esmeralda.cabrera.v@g...'. The main content area shows the 'Overview' page for a deployment named 'nat-p1-20251216234328-43'. The deployment status is 'Complete', indicated by a green checkmark. Key details include the deployment name, subscription ('Azure subscription 1'), resource group ('rg-p1-lab'), start time ('12/16/2025, 11:43:41 PM'), and correlation ID. A 'Go to resource' button is visible. The left sidebar contains links for 'Overview', 'Inputs', 'Outputs', and 'Template'. The right sidebar features recommendations for 'Cost management', 'Microsoft Defender for Cloud', 'Free Microsoft tutorials', and 'Work with an expert'.

Home > nat-p1-20251216234328-43 | Overview

Search resources, services, and docs (G+/)

Microsoft Azure

Search

Delete Cancel Redeploy Download Refresh

✓ Your deployment is complete

Deployment name : nat-p1-20251216234328-43 Start time : 12/16/2025, 11:43:41 PM
Subscription : Azure subscription 1 Correlation ID : 501e81df-5cf5-4137-a7ee-ced...
Resource group : rg-p1-lab

> Deployment details

< Next steps

[Go to resource](#)

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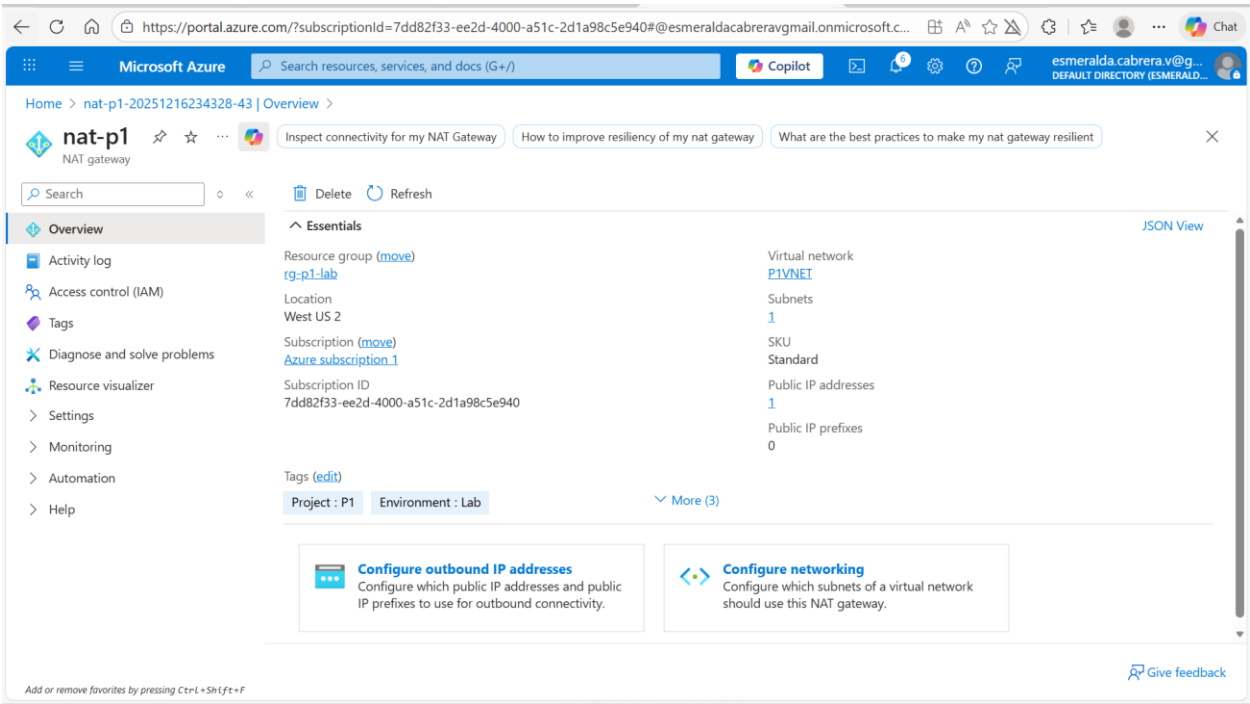
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NAT Gateway Creation with Properties



Step 4: Network Security Groups Configuration

Two NSGs were created:

NSG	Applied To	Allowed Ports
AppNSG	Application Server	22 (SSH), 80 (HTTP)
DbNSG	Database Server	3306 (MySQL), 22 (SSH)

Inbound rules were configured explicitly for these ports to restrict attack surface.

AppNSG Deployment

Home > CreateNetworkSecurityGroupBladeV2-20251217090251 | Overview

Deployment

Search

Delete Cancel Redeploy Download Refresh

Overview

Inputs

Outputs

Template

✓ Your deployment is complete

Deployment name : CreateNetworkSecurityGroupB... Start time : 12/17/2025, 9:04:02 AM
Subscription : Azure subscription 1 Correlation ID : e915269f-bc2f-4b48-b529-55b...
Resource group : rg-p1-lab

> Deployment details

Next steps

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AppNSG Creation

Home > CreateNetworkSecurityGroupBladeV2-20251217090251 | Overview >

AppNSG Network security group

Diagnose connectivity issues related to this security group Retrieve detailed information for troubleshooting security rules +1

Search

Move Delete Refresh Give feedback

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

Settings

Monitoring

Automation

Help

Essentials

Resource group (move) [rg-p1-lab](#)

Location West US 2

Subscription (move) [Azure subscription 1](#)

Subscription ID 7dd82f33-ee2d-4000-a51c-2d1a98c5e940

Tags (edit)

Project : P1 Environment : Lab More (3)

Filter by name

Port == all Protocol == all Source == all Destination == all Action == all

Priority	Name	Port	Protocol	Source	Destination	Action
Inbound Security Rules						
65000	AllowVnetInBound	Any	Any	VirtualNetwork	VirtualNetwork	Allow
65001	AllowAzureLoadBalan...	Any	Any	AzureLoadBalancer	Any	Allow
65500	DenyAllInBound	Any	Any	Any	Any	Deny

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Inbound Security Rules

Home > Network foundation | Network security groups > AppNSG

AppNSG | Inbound security rules

Network security group security rules are evaluated by priority using the combination of source, source port, destination, destination port, and protocol to allow or deny the traffic. A security rule can't have the same priority and direction as an existing rule. You can't delete default security rules, but you can override them with rules that have a higher priority. [Learn more](#)

Filter by name: Port == all Protocol == all Source == all Destination == all Action == all

Priority	Name	Port	Protocol	Source	Destination	Action
100	Allow-SSH	22	TCP	Any	Any	Allow
110	Allow-HTTP	80	TCP	Any	Any	Allow
65000	AllowVnetInBound	Any	Any	VirtualNetwork	VirtualNetwork	Allow
65001	AllowAzureLoadBalan...	Any	Any	AzureLoadBalancer	Any	Allow
65500	DenyAllInBound	Any	Any	Any	Any	Deny

DbNSG Deployment

Home >

CreateNetworkSecurityGroupBladeV2-20251217091606 | Overview

Deployment

Search: Delete Cancel Redeploy Download Refresh

Your deployment is complete

Deployment name : CreateNetworkSecurityGroupB... Start time : 12/17/2025, 9:16:57 AM
Subscription : Azure subscription 1 Correlation ID : 067bb5fa-daeb-47ae-8571-1a...
Resource group : rg-p1-lab

Deployment details

Next steps

[Go to resource](#)

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DbNSG Creation

DbNSG Network security group

Search resources, services, and docs (G+/)

Home > CreateNetworkSecurityGroupBladeV2-20251217091606 | Overview >

Essentials

- Resource group (move) [rg-p1-lab](#)
- Location: West US 2
- Subscription (move) [Azure subscription 1](#)
- Subscription ID: 7dd82f33-ee2d-4000-a51c-2d1a98c5e940
- Tags (edit): Project : P1, Environment : Lab

Custom security rules: 0 inbound, 0 outbound

Associated with: 0 subnets, 0 network interfaces

Filter by name

Port == all Protocol == all Source == all Destination == all Action == all

Priority	Name	Port	Protocol	Source	Destination	Action
Inbound Security Rules						
65000	AllowVnetInBound	Any	Any	VirtualNetwork	VirtualNetwork	Allow
65001	AllowAzureLoadBalanc...	Any	Any	AzureLoadBalancer	Any	Allow
65500	DenyAllInBound	Any	Any	Any	Any	Deny

DbNSG Inbound Rules

DbNSG | Inbound security rules Network security group

Search resources, services, and docs (G+/)

Home > Network foundation | Network security groups > DbNSG

Network security group security rules are evaluated by priority using the combination of source, source port, destination, destination port, and protocol to allow or deny the traffic. A security rule can't have the same priority and direction as an existing rule. You can't delete default security rules, but you can override them with rules that have a higher priority. [Learn more](#)

Filter by name

Port == all Protocol == all Source == all Destination == all Action == all

Priority	Name	Port	Protocol	Source	Destination	Action
100	Allow-SSH	22	TCP	Any	Any	Allow
110	Allow-MySQL	3306	TCP	Any	Any	Allow
65000	AllowVnetInBound	Any	Any	VirtualNetwork	VirtualNetwork	Allow
65001	AllowAzureLoadBalanc...	Any	Any	AzureLoadBalancer	Any	Allow
65500	DenyAllInBound	Any	Any	Any	Any	Deny

Step 5: Application Server Deployment (Public Subnet)

A Linux VM was created to host OwnCloud.

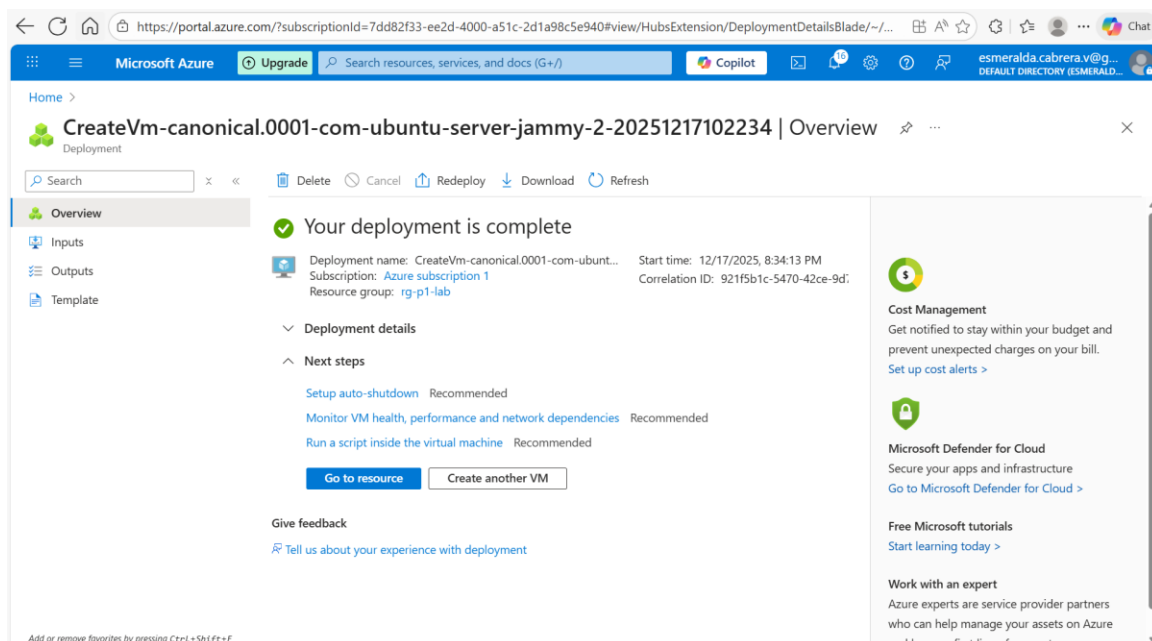
Configuration

- OS: Ubuntu 22.04 LTS
- Size: Standard B-Series VM
- Public IP: Enabled
- NSG: AppNSG
- Authentication: SSH Key

This VM acts as both:

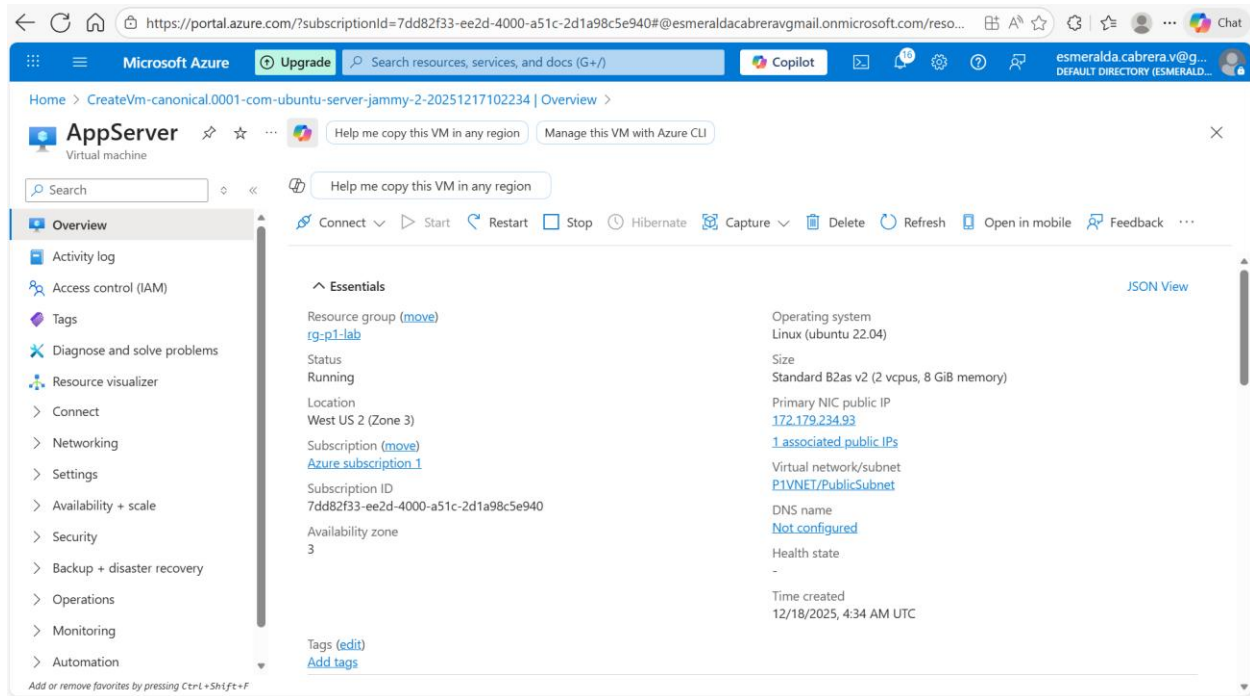
- Web Server
- Bastion Host for database access.

Application Server Deployment



The screenshot displays the Microsoft Azure portal interface. The top navigation bar includes the 'Microsoft Azure' logo, an 'Upgrade' button, a search bar, and a 'Copilot' button. The main header shows the deployment name 'CreateVm-canonical.0001-com-ubuntu-server-jammy-2-20251217102234' and the 'Overview' tab. The left sidebar contains a search bar and a list of tabs: 'Overview', 'Inputs', 'Outputs', and 'Template'. The main content area features a green checkmark icon and the text 'Your deployment is complete'. Below this, deployment details are listed: 'Deployment name: CreateVm-canonical.0001-com-ubuntu...', 'Subscription: Azure subscription 1', and 'Resource group: rg-p1-lab'. The 'Start time' is '12/17/2025, 8:34:13 PM' and the 'Correlation ID' is '921f5b1c-5470-42ce-9d...'. Under 'Deployment details', there are sections for 'Next steps' with links to 'Setup auto-shutdown', 'Monitor VM health, performance and network dependencies', and 'Run a script inside the virtual machine'. At the bottom, there are buttons for 'Go to resource' and 'Create another VM'. On the right side, there are three informational cards: 'Cost Management' (Get notified to stay within your budget and prevent unexpected charges on your bill), 'Microsoft Defender for Cloud' (Secure your apps and infrastructure), and 'Free Microsoft tutorials' (Start learning today). The footer includes a 'Give feedback' link and a note about adding or removing favorites.

Application Server Creation Overview with Properties



Step 6: Database Server Deployment (Private Subnet)

A second VM was deployed inside the private subnet.

Configuration

- OS: Ubuntu
- No Public IP Assigned
- NSG: DbNSG
- Accessible only through SSH from the application server.

This enforces a zero-trust boundary between tiers.

Database Server Deployment

Home > **CreateVm-canonical.0001-com-ubuntu-server-jammy-2-20251217220707** | Overview

Deployment

Search

Delete Cancel Redeploy Download Refresh

Overview

- Inputs
- Outputs
- Template

Your deployment is complete

Deployment name: CreateVm-canonical.0001-com-ubuntu-server-jammy-2-20251217220707 Start time: 12/17/2025, 10:11:29 PM
Subscription: [Azure subscription 1](#) Correlation ID: e4b7a2ab-4cfa-4084-a97
Resource group: [rg-p1-lab](#)

Deployment details

Next steps

- [Setup auto-shutdown](#) Recommended
- [Monitor VM health, performance and network dependencies](#) Recommended
- [Run a script inside the virtual machine](#) Recommended

[Go to resource](#) [Create another VM](#)

Give feedback

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Database Server Creation with Properties

Home > **CreateVm-canonical.0001-com-ubuntu-server-jammy-2-20251217220707** | Overview > **DbServer**

Virtual machine

Search

Overview

- Activity log
- Access control (IAM)
- Tags
- Diagnose and solve problems
- Resource visualizer
- Connect
 - Connect
- Networking
 - Bastion
- Settings
 - Availability + scale
 - Security
 - Backup + disaster recovery
 - Operations

Essentials

Resource group [\(move\)](#) [rg-p1-lab](#)

Status: Running

Location: West US 2 (Zone 3)

Subscription [\(move\)](#) [Azure subscription 1](#)

Subscription ID: 7dd82f33-ee2d-4000-a51c-2d1a98c5e940

Availability zone: 3

Operating system: Linux (ubuntu 22.04)

Size: Standard B2as v2 (2 vcpus, 8 GiB memory)

Primary NIC public IP: [1 associated public IPs](#)

Virtual network/subnet: [P1VNET/PrivateSubnet](#)

DNS name: -

Health state: -

Time created: 12/18/2025, 6:11 AM UTC

Step 7: Secure SSH Access Workflow

To reach the private DB server:

1. Copied database key to application VM using SCP.
2. SSH'd into application VM.
3. From there, SSH'd into the private database VM.

This simulates a bastion-host architecture.

Step 8: MySQL Installation (Database Server)

MySQL was installed via automated script:

```
sudo apt update
```

```
wget <installation script>
```

```
chmod 700 script
```

```
dos2unix script
```

```
./script
```

This configured the database backend required by OwnCloud.

Executing the Script

```
ubuntu@DbServer: ~  
  
The programs included with the Ubuntu system are free software;  
the exact distribution terms for each program are described in the  
individual files in /usr/share/doc/*/copyright.  
  
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by  
applicable law.  
  
To run a command as administrator (user "root"), use "sudo <command>".  
See "man sudo_root" for details.  
  
ubuntu@DbServer:~$ sudo apt update  
wget https://d60pu47qoi4ee.cloudfront.net/azure_install_mysql.sh  
sudo chmod 700 azure_install_mysql.sh  
sudo apt install dos2unix  
sudo dos2unix ./azure_install_mysql.sh  
sudo ./azure_install_mysql.sh  
Hit:1 http://azure.archive.ubuntu.com/ubuntu jammy InRelease  
Hit:2 http://azure.archive.ubuntu.com/ubuntu jammy-updates InRelease  
Hit:3 http://azure.archive.ubuntu.com/ubuntu jammy-backports InRelease  
Hit:4 http://azure.archive.ubuntu.com/ubuntu jammy-security InRelease  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
All packages are up to date.  
--2025-12-18 06:44:49-- https://d60pu47qoi4ee.cloudfront.net/azure_install_mysql.sh  
Resolving d60pu47qoi4ee.cloudfront.net (d60pu47qoi4ee.cloudfront.net)... 52.85.97.216, 52.85.97.28, 52.85.97.148, ...  
Connecting to d60pu47qoi4ee.cloudfront.net (d60pu47qoi4ee.cloudfront.net)|52.85.97.216|:443... connected.  
HTTP request sent, awaiting response... 200 OK  
Length: 1339 (1.3K) [text/x-sh]  
Saving to: 'azure_install_mysql.sh'  
  
azure_install_mysql.sh      100%[=====] 1.31K --.-KB/s  in 0s  
  
2025-12-18 06:44:49 (777 MB/s) - 'azure_install_mysql.sh' saved [1339/1339]
```

Downloading the Script

```
ubuntu@DbServer: ~  
reading /usr/share/mecab/dic/ipadic/matrix.def ... 1316x1316  
emitting matrix      : 100% |#####|  
  
done!  
update-alternatives: using /var/lib/mecab/dic/ipadic-utf8 to provide /var/lib/mecab/dic/debian (mecab-dictionary) in auto mode  
Setting up libhtml-parser-perl:amd64 (3.76-1build2) ...  
Setting up libhttp-message-perl (6.36-1) ...  
Setting up mysql-server-8.0 (8.0.44-0ubuntu0.22.04.2) ...  
update-alternatives: using /etc/mysql/mysql.cnf to provide /etc/mysql/my.cnf (my.cnf) in auto mode  
Renaming removed key_buffer and myisam-recover options (if present)  
mysqld will log errors to /var/log/mysql/error.log  
mysqld is running as pid 6588  
Created symlink /etc/systemd/system/multi-user.target.wants/mysql.service → /lib/systemd/system/mysql.service.  
Setting up libcgi-pm-perl (4.54-1) ...  
Setting up libhtml-template-perl (2.97-1.1) ...  
Setting up mysql-server (8.0.44-0ubuntu0.22.04.2) ...  
Setting up libcgi-fast-perl (1:2.15-1) ...  
Processing triggers for man-db (2.10.2-1) ...  
Processing triggers for libc-bin (2.35-0ubuntu3.11) ...  
Scanning processes...  
Scanning linux images...  
  
Running kernel seems to be up-to-date.  
  
No services need to be restarted.  
  
No containers need to be restarted.  
  
No user sessions are running outdated binaries.  
  
No VM guests are running outdated hypervisor (qemu) binaries on this host.  
Installed MySQL  
Configuring MySQL now  
MySQL Configuration complete  
bind-address = 0.0.0.0  
mysqlx-bind-address = 0.0.0.0  
ubuntu@DbServer:~$
```

Step 9: OwnCloud Installation (Application Server)

OwnCloud and dependencies were installed using Apache, PHP, and required modules.

Key commands executed:

```
sudo apt install apache2 php7.4 ...
```

```
wget owncloud package
```

```
tar -xjf package
```

```
chown -R www-data owncloud
```

```
systemctl restart apache2
```

The application was deployed under /var/www/html/owncloud.

Running the initial Update Script Part 1

```
ubuntu@AppServer: ~  
logout  
Connection to 10.0.2.5 closed.  
ubuntu@AppServer:~$ sudo apt update  
sudo add-apt-repository ppa:ondrej/php -y  
sudo apt update  
  
Hit:1 http://azure.archive.ubuntu.com/ubuntu jammy InRelease  
Hit:2 http://azure.archive.ubuntu.com/ubuntu jammy-updates InRelease  
Hit:3 http://azure.archive.ubuntu.com/ubuntu jammy-backports InRelease  
Hit:4 http://azure.archive.ubuntu.com/ubuntu jammy-security InRelease  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
All packages are up to date.  
PPA publishes dbgsym, you may need to include 'main/debug' component  
Repository: 'deb https://ppa.launchpadcontent.net/ondrej/php/ubuntu/ jammy main'  
Description:  
Co-installable PHP versions: PHP 5.6, PHP 7.x, PHP 8.x and most requested extensions are included. Packages are provided for *Current  
* Ubuntu *LTS* releases (https://wiki.ubuntu.com/Releases). Expanded Security Maintenance releases ARE NOT supported.  
  
Debian stable, oldstable and Debian LTS packages are provided from a separate repository: https://deb.sury.org/#debian-dpa  
  
You can get more information about the packages at https://deb.sury.org  
  
BUGS&FEATURES: This PPA has a issue tracker:  
https://deb.sury.org/#bug-reporting  
  
Issues reported in a private email don't scale and most likely will be ignored. I simply don't have capacity to answer questions pri  
vately.  
  
CAVEATS:  
1. If you are using apache2, you are advised to add ppa:ondrej/apache2  
2. If you are using nginx, you are advised to add ppa:ondrej/nginx  
  
DONATION: If you like my work and you want to show appreciation, please consider donating regularly: https://donate.sury.org/  
  
COMMERCIAL SUPPORT: Support for PHP packages for older Debian and Ubuntu release can be bought from https://www.freexian.com/lts/php/
```

Running the initial Update Script Part 2

```
ubuntu@AppServer: ~
BUGS&FEATURES: This PPA has a issue tracker:
https://deb.sury.org/#bug-reporting

Issues reported in a private email don't scale and most likely will be ignored. I simply don't have capacity to answer questions privately.

CAVEATS:
1. If you are using apache2, you are advised to add ppa:ondrej/apache2
2. If you are using nginx, you are advised to add ppa:ondrej/nginx

DONATION: If you like my work and you want to show appreciation, please consider donating regularly: https://donate.sury.org/

COMMERCIAL SUPPORT: Support for PHP packages for older Debian and Ubuntu release can be bought from https://www.freexian.com/lts/php/
More info: https://launchpad.net/~ondrej/+archive/ubuntu/php
Adding repository.
Adding deb entry to /etc/apt/sources.list.d/ondrej-ubuntu-php-jammy.list
Adding disabled deb-src entry to /etc/apt/sources.list.d/ondrej-ubuntu-php-jammy.list
Adding key to /etc/apt/trusted.gpg.d/ondrej-ubuntu-php.gpg with fingerprint B8DC7E5394656EFBCE4C1DD71DAEAAB4AD4CAB6
Hit:1 http://azure.archive.ubuntu.com/ubuntu jammy InRelease
Hit:2 http://azure.archive.ubuntu.com/ubuntu jammy-updates InRelease
Hit:3 http://azure.archive.ubuntu.com/ubuntu jammy-backports InRelease
Hit:4 http://azure.archive.ubuntu.com/ubuntu jammy-security InRelease
Get:5 https://ppa.launchpadcontent.net/ondrej/php/ubuntu jammy InRelease [24.6 kB]
Get:6 https://ppa.launchpadcontent.net/ondrej/php/ubuntu jammy/main amd64 Packages [147 kB]
Get:7 https://ppa.launchpadcontent.net/ondrej/php/ubuntu jammy/main Translation-en [46.3 kB]
Fetched 218 kB in 2s (115 kB/s)
Reading package lists... Done
Hit:1 http://azure.archive.ubuntu.com/ubuntu jammy InRelease
Hit:2 http://azure.archive.ubuntu.com/ubuntu jammy-updates InRelease
Hit:3 http://azure.archive.ubuntu.com/ubuntu jammy-backports InRelease
Hit:4 http://azure.archive.ubuntu.com/ubuntu jammy-security InRelease
Hit:5 https://ppa.launchpadcontent.net/ondrej/php/ubuntu jammy InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
3 packages can be upgraded. Run 'apt list --upgradable' to see them.
ubuntu@AppServer:~$
```

Executing the Apache Installs Script

```
ubuntu@AppServer: ~
ubuntu@AppServer:~$ sudo apt install -y apache2 libapache2-mod-php7.4 mariadb-server openssl redis-server wget php7.4 php7.4-imagick
php7.4-common php7.4-curl php7.4-gd php7.4-imap php7.4-intl php7.4-json php7.4-mbstring php7.4-gmp php7.4-bcmath php7.4-mysql php7.4-
ssh2 php7.4-xml php7.4-zip php7.4-apcu php7.4-redis php7.4-ldap php-phpseclib
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
openssl is already the newest version (3.0.2-0ubuntu1.20).
openssl set to manually installed.
wget is already the newest version (1.21.2-2ubuntu1.1).
wget set to manually installed.
The following additional packages will be installed:
 apache2-bin apache2-data apache2-utils bzip2 fontconfig-config fonts-dejavu-core fonts-droid-fallback fonts-noto-mono
 fonts-urw-base35 galera-4 ghostscript gsfonts imagemagick-6-common libabsl20210324 libaom3 libapr1 libaprutil1
 libaprutil1-dbd-sqlite3 libaprutil1-ldap libatomic1 libavahi-client3 libavahi-common-data libavahi-common3 libavif13
 libc-client2007e libcgi-fast-perl libcgi-pm-perl libclone-perl libconfig-inifiles-perl libcupsl2 libdavid5 libdbd-mysql-perl
 libdbi-perl libde265-0 libdeflate0 libencode-locale-perl libfcgi-bin libfcgi-perl libfcgi0ldbl libfftw3-double3 libfontconfig1
 libgavl-0 libgd3 libgomp1 libgs9 libgs9-common libheif1 libhtml-parser-perl libhtml-tagset-perl libhtml-template-perl
 libhttp-date-perl libhttp-message-perl libidn12 libijs-0.35 libimagequant0 libio-html-perl libjbig0 libjbig2dec0 libjemalloc2
 libjpeg-turbo8 libjpeg8 liblcms2-2 liblqr-1-0 libltdl7 liblua5.1-0 liblua5.3-0 liblwp-mediatypes-perl liblzfl
 libmagickcore-6.q16-6 libmagickwand-6.q16-6 libmariadb3 libmysqlclient21 libonig5 libopenjp2-7 libpaper-utils libpaper1
 libsnappy1v5 libssh2-1 libtiff5 libtimedate-perl liburi-perl liburing2 libwebp7 libwebpdemux2 libwebpmux3 libx265-199 libxpm4
 libyuv0 libzip4 lua-bitop lua-cjson mailcap mariadb-client-10.6 mariadb-client-core-10.6 mariadb-common mariadb-server-10.6
 mariadb-server-core-10.6 mime-support mlock mysql-common php-common php7.4-cli php7.4-igbinary php7.4-opcache php7.4-readline
 poppler-data redis-tools socat ssl-cert
Suggested packages:
 apache2-doc apache2-suexec-pristine | apache2-suexec-custom www-browser bzip2-doc fonts-noto fonts-freefont-otf
 | fonts-freefont-ttf fonts-texgyre ghostscript-x php-pear uw-mailutils cups-common libmldbm-perl libnet-daemon-perl
 libsql-statement-perl libfftw3-bin libfftw3-dev libgd-tools libdata-dump-perl libipc-sharedcache-perl liblcms2-utils
 libmagickcore-6.q16-6-extra libbusiness-isbn-perl libwww-perl mailx mariadb-test php-libsodium php-mcrypt poppler-utils
 fonts-japanese-mincho | fonts-ipafont-mincho fonts-japanese-gothic | fonts-ipafont-gothic fonts-arphic-ukai fonts-arphic-uming
 fonts-nanum ruby-redis
The following NEW packages will be installed:
 apache2 apache2-bin apache2-data apache2-utils bzip2 fontconfig-config fonts-dejavu-core fonts-droid-fallback fonts-noto-mono
 fonts-urw-base35 galera-4 ghostscript gsfonts imagemagick-6-common libabsl20210324 libaom3 libapache2-mod-php7.4 libapr1
 libaprutil1 libaprutil1-dbd-sqlite3 libaprutil1-ldap libatomic1 libavahi-client3 libavahi-common-data libavahi-common3 libavif13
 libc-client2007e libcgi-fast-perl libcgi-pm-perl libclone-perl libconfig-inifiles-perl libcupsl2 libdavid5 libdbd-mysql-perl
 libdbi-perl libde265-0 libdeflate0 libencode-locale-perl libfcgi-bin libfcgi-perl libfcgi0ldbl libfftw3-double3 libfontconfig1
```

End of Apache Installs Download

```
ubuntu@AppServer: ~  
Enabling conf other-vhosts-access-log.  
Enabling conf security.  
Enabling conf serve-cgi-bin.  
Enabling site 000-default.  
info: Switch to mpm prefork for package libapache2-mod-php7.4  
Module mpm_event disabled.  
Enabling module mpm_prefork.  
info: Executing deferred 'a2enmod php7.4' for package libapache2-mod-php7.4  
Enabling module php7.4.  
Created symlink /etc/systemd/system/multi-user.target.wants/apache2.service → /lib/systemd/system/apache2.service.  
Created symlink /etc/systemd/system/multi-user.target.wants/apache-htcacheclean.service → /lib/systemd/system/apache-htcacheclean.service.  
Setting up ghostscript (9.55.0~dfsg1-0ubuntu5.13) ...  
Setting up mariadb-server (1:10.6.22-0ubuntu0.22.04.1) ...  
Setting up php7.4 (1:7.4.33-21+ubuntu22.04.1+deb.sury.org+1) ...  
Setting up libgd3:amd64 (2.3.3-13+ubuntu22.04.1+deb.sury.org+1) ...  
Setting up php7.4-gd (1:7.4.33-21+ubuntu22.04.1+deb.sury.org+1) ...  
  
Creating config file /etc/php/7.4/mods-available/gd.ini with new version  
Processing triggers for ufw (0.36.1-4ubuntu0.1) ...  
Processing triggers for man-db (2.10.2-1) ...  
Processing triggers for libc-bin (2.35-0ubuntu3.11) ...  
Processing triggers for php7.4-cli (1:7.4.33-21+ubuntu22.04.1+deb.sury.org+1) ...  
Processing triggers for libapache2-mod-php7.4 (1:7.4.33-21+ubuntu22.04.1+deb.sury.org+1) ...  
Scanning processes...  
Scanning linux images...  
  
Running kernel seems to be up-to-date.  
  
No services need to be restarted.  
  
No containers need to be restarted.  
  
No user sessions are running outdated binaries.  
  
No VM guests are running outdated hypervisor (qemu) binaries on this host.  
ubuntu@AppServer:~$
```

Owncloud Download Complete

```
ubuntu@AppServer: /var/www$  
Running kernel seems to be up-to-date.  
  
No services need to be restarted.  
  
No containers need to be restarted.  
  
No user sessions are running outdated binaries.  
  
No VM guests are running outdated hypervisor (qemu) binaries on this host.  
ubuntu@AppServer:~$ sudo a2enmod dir env headers mime rewrite setenvif  
sudo systemctl restart apache2  
cd /var/www/html  
sudo rm *  
sudo wget https://download.owncloud.com/server/stable/owncloud-complete-latest.tar.bz2  
sudo tar -xjf owncloud-complete-latest.tar.bz2  
sudo chown -R www-data: owncloud  
sudo systemctl restart apache2  
Module dir already enabled  
Module env already enabled  
Enabling module headers.  
Module mime already enabled  
Enabling module rewrite.  
Module setenvif already enabled  
To activate the new configuration, you need to run:  
  systemctl restart apache2  
--2025-12-18 06:58:11-- https://download.owncloud.com/server/stable/owncloud-complete-latest.tar.bz2  
Resolving download.owncloud.com (download.owncloud.com)... 167.233.14.167, 2a01:4f8:1c1d:3d1::1  
Connecting to download.owncloud.com (download.owncloud.com)|167.233.14.167|:443... connected.  
HTTP request sent, awaiting response... 200 OK  
Length: 75611691 (72M) [application/x-bzip2]  
Saving to: 'owncloud-complete-latest.tar.bz2'  
  
owncloud-complete-latest.tar.bz2 100%[=====>] 72.11M 20.7MB/s in 4.4s  
  
2025-12-18 06:58:16 (16.4 MB/s) - 'owncloud-complete-latest.tar.bz2' saved [75611691/75611691]  
ubuntu@AppServer:/var/www/html$
```

Step 10: Validation Testing

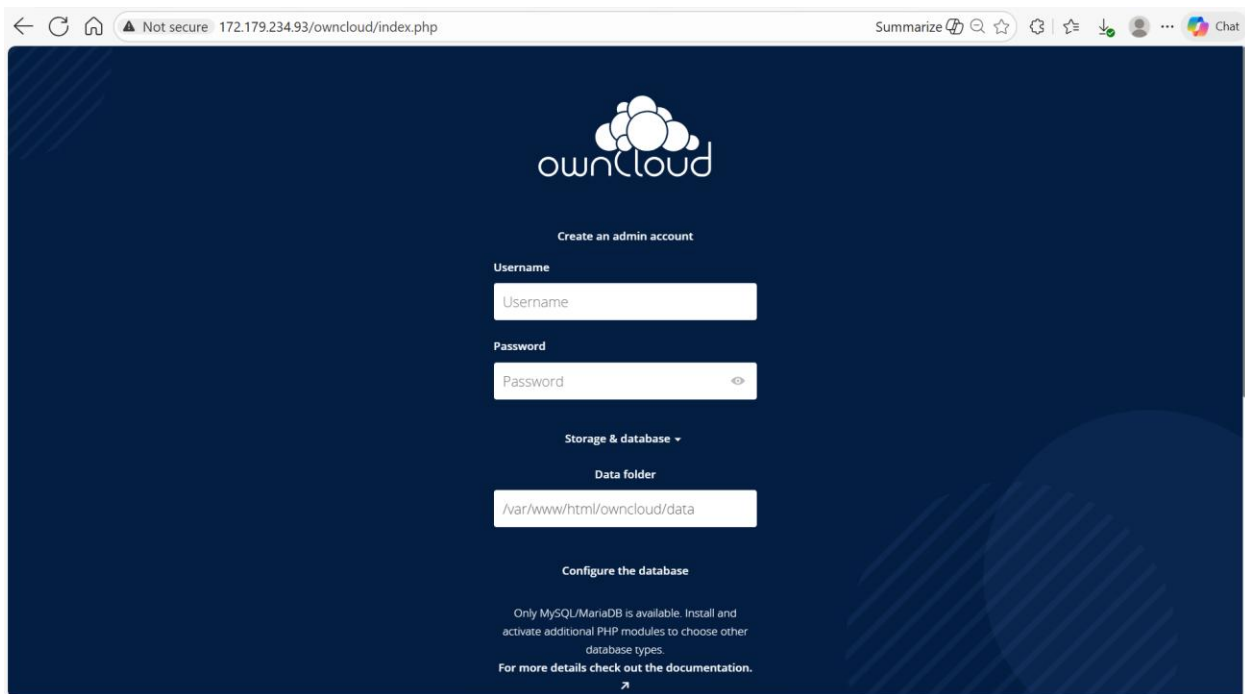
The deployment was validated by navigating to:

`http://<Application-Public-IP>/owncloud`

Successful page load confirmed:

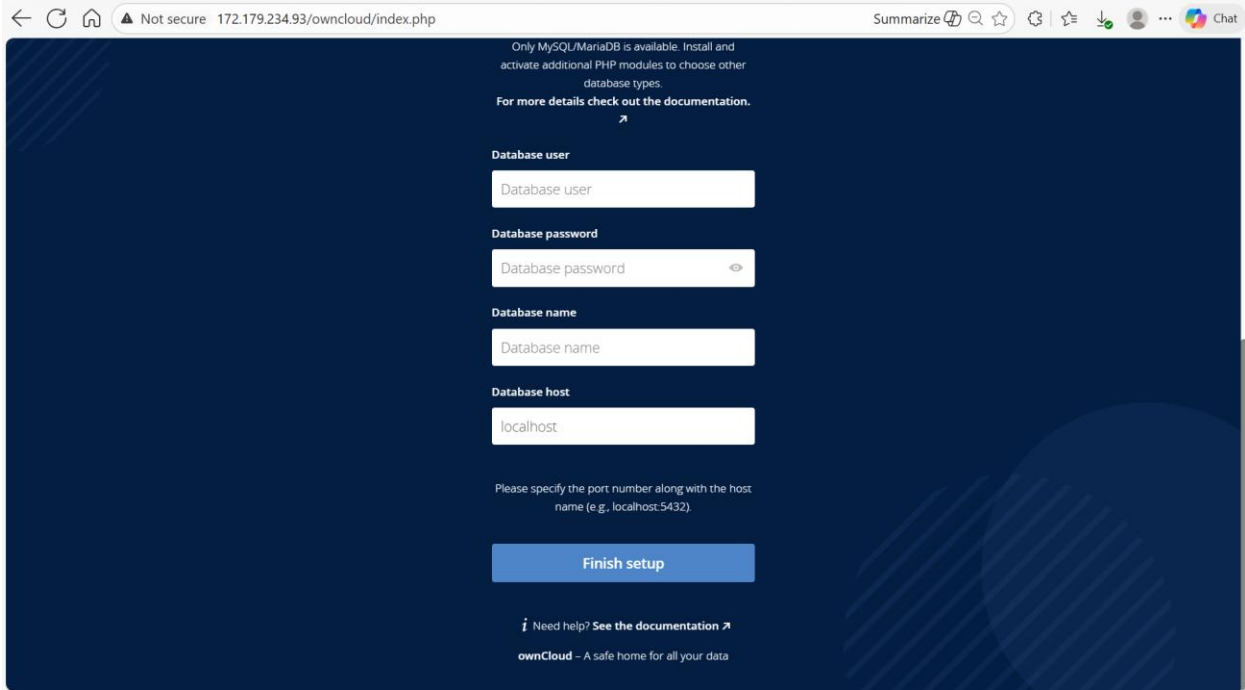
- Web tier reachable
- Database connectivity functional
- Secure architecture operating as designed.

Accessing the Owncloud Application Via the Web Server. Part 1 of Access Portal.



The screenshot shows a web browser window with the URL `172.179.234.93/owncloud/index.php`. The page has a dark blue background with the Owncloud logo at the top center. Below the logo, there is a section titled "Create an admin account" with two input fields: "Username" and "Password". The "Password" field has a toggle icon (an eye) to the right. Below these fields, there is a section titled "Storage & database" with a dropdown menu. Under the dropdown, there is a "Data folder" label and an input field containing the path `/var/www/html/owncloud/data`. At the bottom, there is a section titled "Configure the database" with a note: "Only MySQL/MariaDB is available. Install and activate additional PHP modules to choose other database types." and a link: "For more details check out the documentation." with an external link icon.

Accessing the Owncloud Application Via the Web Server. Part 2 of Access Portal.



The screenshot shows a web browser window with the URL `172.179.234.93/owncloud/index.php`. The page has a dark blue background with a light blue diagonal stripe pattern. At the top, a message states: "Only MySQL/MariaDB is available. Install and activate additional PHP modules to choose other database types. For more details check out the documentation." Below this, there are four input fields: "Database user" (containing "Database user"), "Database password" (containing "Database password" with a toggle icon), "Database name" (containing "Database name"), and "Database host" (containing "localhost"). A note below the host field says: "Please specify the port number along with the host name (e.g., localhost:5432)." A blue "Finish setup" button is centered below the fields. At the bottom, there is a link "Need help? See the documentation" and the footer "ownCloud - A safe home for all your data".

Security Design Principles Applied

This project demonstrates several enterprise cloud security patterns:

- **Network Segmentation** – Public vs Private Subnets
- **No Public DB Exposure**
- **Controlled Egress via NAT**
- **Least Privilege NSG Rules**
- **SSH Key Authentication**
- **Bastion-style Access Flow**

Cost Awareness and Cleanup

Cloud operates on a pay-per-use model, so all resources must be deleted after completion to avoid charges.

Key Learning Outcomes

This project demonstrates practical ability to:

- Design secure Azure networking.
- Deploy multi-tier infrastructure.
- Configure Linux workloads in cloud environments.
- Implement self-hosted SaaS replacement solutions.
- Apply enterprise security controls in IaaS deployments.

Future Enhancements

- Replace VM-based MySQL with Azure Database for MySQL (PaaS).
- Add Azure Application Gateway + WAF.
- Implement Azure AD authentication.
- Enable TLS encryption with managed certificates.
- Add backup and monitoring via Azure Monitor.